Tutorial of Medicinal Chemistry Filters (MCFs) and MCE-18 Descriptor Calculation by MolAICal

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1. Introduction

Medicinal chemistry filters (MCFs) function to eliminate compounds that contain so-called "**structural alerts**"[1]. Molecules with such moieties either have **unstable or reactive groups,** or they undergo biotransformations that result in the production of toxic metabolites or intermediates. PAINS (pan-assay interfering compounds) filters consist of a set of substructure filters designed to reduce false positives, assay artifacts, and non-specific bioactive molecules in screening libraries. They are based on the premise that certain fragments within a structure can give rise to undesirable properties—such as reactivity, chelation, colloidal aggregate formation, and dye-related effects—that interfere with assay results. Since both MCFs and PAINS involve reactivity, here, **MCFs** function in MolAICal **encompasses structures covered by both MCFs and PAINS.**

MCE-18[2], standing for "medicinal chemistry evolution, 2018", is an original molecular descriptor proposed to effectively score the novelty and lead potential of pharmacologically relevant molecules. **Formula of MCE-18:**

$$\text{MCE-18}=\left(\text{AR}+\text{NAR}+\text{CHIRAL}+\text{SPIRO}+\frac{\overbrace{\text{sp}^3+\text{Cyc}-\text{Acyc}}^{\text{NCSPTR}}}{1+\text{sp}^3}\right)\times \text{Q}^1$$

Explanation of Parameters

- ✧ **AR**: Presence of an aromatic or heteroaromatic ring (0 = absent, 1 = present).
- ✧ **NAR**: Presence of an aliphatic or heteroaliphatic ring (0 = absent, 1 = present).
- ✧ **CHIRAL**: Presence of a chiral center (0 = absent, 1 = present).
- ✧ **SPIRO**: Presence of a *spiro* point (0 = absent, 1 = present).
- ✧ **sp^3**: Proportion of sp^3-hybridized carbon atoms (ranging from 0 to 1).
- ✧ **Cyc**: Proportion of cyclic carbons that are sp^3-hybridized (ranging from 0 to 1).
- ✧ **Acyc**: Proportion of acyclic carbons that are sp^3-hybridized (ranging from 0 to 1).
- ✧ **NCSPTR**: A composite term calculated as (sp^3 + Cyc - Acyc).
- ✧ **Q^1**: Normalized quadratic index, encoding the degree of branching of the structural framework.

For more detailed MolAICal, please visit: https://molaical.github.io.

2. Materials

2.1. Software requirements

1) MolAICal: https://molaical.github.io or https://molaical.gitlab.io

Notice: Make sure the installation of MolAICal is correct!

2.2. Example files

1) All the necessary tutorial files are downloaded from:
https://github.com/MolAICal/tutorials/tree/master/024-mcf_mce18

Part I. Calculation of Chemistry Filters (MCFs) by MolAICal

1) Calculate MCFs for a single SMILES string

```
#> molaical.exe -tool mcf -s "CC(=O)OC1=CC=CC=C1C(=O)O"
```

2) Calculate MCFs for a single SMILES string without properties

```
#> molaical.exe -tool mcf -s "CC(=O)OC1=CC=CC=C1C(=O)O" -p false
```

3) Calculate MCFs with output file and use format (e.g., 'text', 'json', or 'csv')

```
#> molaical.exe -tool mcf -s "CC(=O)OC1=CC=CC=C1C(=O)O" -o result.csv -f csv
```

4) Calculate MCFs with the output file and no properties output

```
#> molaical.exe -tool mcf -s "CC(=O)OC1=CC=CC=C1C(=O)O" -o result.csv -f csv -p false
```

Here, some properties are also output by default for further assessment of molecules:

- **SMILES**: the SMILES sequence of the input molecule.
- **Valid**: determines whether a molecule is valid; its value is True or False.
- **Passes filters**: determines whether it passes the MCFs evaluation, its value is True or False. If the value is True, it indicates that the molecule passes the MCFs and PAINS filters
- **LogP**: octanol-water partition coefficient, a proportion of a substance's concentration in the octanol phase relative to its concentration in the aqueous phase of a biphasic octanol/water system.
- **SA Score**: also known as **Synthetic Accessibility Score**, a rule-of-thumb assessment of the **difficulty (rated 10)** or **ease (rated 1)** of synthesizing a specific molecule. The SA score is derived from a blend of contributions from the molecule's fragments[3].
- **NP Score**: also known as **Natural Product-likeness Score**, a Bayesian metric that enables the assessment of how similar molecules are to the structural space encompassed by natural products[4].
- **QED**: also known as **Quantitative Estimation of Drug-likeness**, a [0, 1] value assessing the likelihood of a molecule being a feasible drug candidate. Like SA, the descriptor thresholds in QED have shifted over the past decade, and current thresholds may not include the most recent drugs[5].
- **Molecular weight (MW)**: the summation of atomic weights in a compound.

Download "024-mcf_mce18/molecules.smi" from the above URL; "molecules.smi" can contain many SMILES strings, so MolAICal can **calculate MCFs for one or many molecules at one time.** Please try the below examples:

1) Calculate MCFs from the file, which can contain multiple lines of SMILES, with one molecule's SMILES represented per line

```
#> molaical.exe -tool mcf -i molecules.smi -o result.csv -f csv
```

2) Calculate MCFs from the file without properties output, which can contain multiple lines of SMILES, with one molecule's SMILES represented per line.
#> molaical.exe -tool mcf -i molecules.smi -o result.csv -f csv -p false

Note: Users can select one of the above ways to calculate MCFs. For more detailed help, please manual of MolAICal.

Part II. Calculation of MCE-18 by MolAICal

1) Calculate MCE-18 for a single SMILES
#> molaical.exe -tool mce18 -s "CC(=O)OC1=CC=CC=C1C(=O)O"

2) Only output SMILES and MCE-18 value
#> molaical.exe -tool mce18 -s "CC(=O)OC1=CC=CC=C1C(=O)O" -sm true

3) Input SMILES and output CSV
#> molaical.exe -tool mce18 -s "CC(=O)OC1=CC=CC=C1C(=O)O" -o output.csv

4) Input SMILES and output CSV without detailed output
#> molaical.exe -tool mce18 -s "CC(=O)OC1=CC=CC=C1C(=O)O" -o output.csv -q true

5) Input SMILES and output CSV only containing SMILES and MCE-18 value
#> molaical.exe -tool mce18 -s "CC(=O)OC1=CC=CC=C1C(=O)O" -o output.csv -sm true

6) Input file containing SMILES strings (one per line) and output CSV file
#> molaical.exe -tool mce18 -i molecules.smi -o results.csv

7) Input file containing SMILES strings (one per line) and output CSV only containing SMILES and MCE-18
#> molaical.exe -tool mce18 -i molecules.smi -o results.csv -sm true

The analysis for MCE-18 values:

➢ **From practical application scenarios, 45-78 is a relatively optimal range**

Analysis of supplier compound libraries (e.g., ChemDiv, Enamine) shows[2]:

- MCE-18 < 45: Molecules have simple structures, low novelty, mostly "old scaffolds", and limited attractiveness;
- MCE-18: 45-78: Sufficient novelty, in line with current trends in medicinal chemistry (45-63), and high structural similarity to recent patent compounds of pharmaceutical companies (63−78);
- MCE-18 > 78: Need manual evaluation of target adaptability and drug-likeness, as excessive complexity may lead to development difficulties.

Note: Users can select one of the above ways to calculate MCE-18. For more detailed help, please manual of MolAICal.

"Only the calculations for MCFs and MCE-18 need to be studied; reading up to this point is sufficient. The following content is optional and may be useful when converting molecules from other formats to canonical SMILES or performing batch canonical SMILES conversions."

Part III. Convert various formats to canonical SMILES

For the purposes of calculating MCE-18, SA, and MCFs, chemical file formats such as mol2, sdf, mol, etc., often need to be converted into canonical SMILES notation. In this context, MolAICal offers an RDKIT-based method for converting molecular structures from formats including mol2, sdf, mol, pdb, inchi, and SMILES into their canonical SMILES representations.

1) Mol2 to canonical SMILES

```
#> molaical.exe -tool tosmi -i lig.mol2 -t mol2
```

2) Mol2 to canonical SMILES without conversion failure messages

```
#> molaical.exe -tool tosmi -i lig.mol2 -t mol2 -q true
```

Note: -q: values: true and false, "true" means to suppress conversion failure messages.

3) SMILES to canonical SMILES

```
#> molaical.exe -tool tosmi -i "CCO" -t smiles
```

4) InChI to canonical SMILES

```
#> molaical.exe -tool tosmi -i "InChI=1S/C2H6O/c1-2-3/h3H,2H2,1H3" -t inchi
```

5) PDB to canonical SMILES

```
#> molaical.exe -tool tosmi -i lig.pdb -t pdb
```

6) mol to canonical SMILES

```
#> molaical.exe -tool tosmi -i lig.mol -t mol
```

7) SDF to canonical SMILES

```
#> molaical.exe -tool tosmi -i lig.sdf -t sdf
```

Note: "-q, -u, or -r" can also enable the conversion of molecular formats such as .mol2, .sdf, .mol, .pdb, InChI, SMILES to canonical SMILES.

Part IV. Batch-converted to SMILES strings

Sometimes, it is necessary to compute molecular files in formats such as .mol2, .sdf, .mol, .pdb, and InChI for MCFs and MCE-18 descriptors. These files should be batch-converted to SMILES strings to enable calculation using the aforementioned method.

Here, two Linux bash scripts are provided for batch-format conversion:

1) Open **“024-mcf_mce18\batch”**, and check **run_combine.sh:**

```
#!/bin/bash
cp /dev/null all.smi
for i in `ls -l | grep ".sdf" | awk '{print $9}'`
do
    # Above ".sdf" is a keyword. The below command is to remove the suffix.
    molaical.exe -tool tosmi -t sdf -q true -i $i > tmp.smi
    # Output last line.
        tail -n 1 tmp.smi >> all.smi
done
rm tmp.smi
```

Users can rename “all.smi” which contains all SMILES strings from example files in SDF format.

2) Open **“024-mcf_mce18\batch”**, and check **run_individual.sh:**

```
#!/bin/bash
for i in `ls -l | grep ".sdf" | awk '{print $9}'`
do
    # Above ".sdf" is a keyword. The below command is to remove the suffix.
    tmp=$(echo $i | awk -F'.' '{print $1}')
    # echo $tmp
    molaical.exe -tool tosmi -t sdf -q true -i $i > tmp.smi
    # Output last line.
    tail -n 1 tmp.smi > $tmp'.smi'
done
rm tmp.smi
```

This script will convert an SDF file to a file containing a SMILES string one by one. The SDF file has the same suffix name with the corresponding converted file that contains a SMILES string.

3) Run **run_combine.sh** and **run_individual.sh**

Before running the scripts, the path of "molaical.exe" should be given correctly. In the Linux console:
For the first script:

```
#> chmod +x run_combine.sh
#> ./run_combine.sh
```

For the second script:

```
#> chmod +x run_individual.sh
#> ./run_individual.sh
```

Note: the above example is based on molecules in SDF format, and the same method is also applicable to molecules in mol2, sdf, mol, pdb, inchi, and SMILES and other formats. Please modify the letters such as ".sdf" to ".mol2", etc, in the corresponding scripts.

Reference

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2. Ivanenkov, Y. A.; Zagribelnyy, B. A.; Aladinskiy, V. A., Are We Opening the Door to a New Era of Medicinal Chemistry or Being Collapsed to a Chemical Singularity? *Journal of Medicinal Chemistry* **2019,** *62* (22), 10026-10043.
3. Ertl, P.; Schuffenhauer, A., Estimation of synthetic accessibility score of drug-like molecules based on molecular complexity and fragment contributions. *J Cheminform* **2009,** *1* (1), 8.
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5. Shultz, M. D., Two Decades under the Influence of the Rule of Five and the Changing Properties of Approved Oral Drugs. *J Med Chem* **2019,** *62* (4), 1701-1714.